

Joint Optimization of VNF Placement and Proactive Migration under Data-Plane Hardware Constraints

Hoàng Duy Hưng

Supervisor: Assoc. Prof. Nguyễn Ngọc Hoà

Ngày 14 tháng 4 năm 2026

Tóm tắt nội dung

Service Function Chaining (SFC) in Network Function Virtualization (NFV) and Software-Defined Networking (SDN) environments enables the dynamic composition of ordered Virtual Network Functions (VNFs) to deliver complex network services. However, practical SFC deployments in modern data center networks (DCNs) face significant challenges, including end-to-end latency accumulation, resource fragmentation, and high migration overhead, especially under dynamic traffic conditions such as sudden surges or distributed denial-of-service (DDoS) attacks. The adoption of programmable data planes based on Segment Routing over IPv6 (SRv6) and P4-programmable switches introduces powerful traffic-steering capabilities but imposes strict hardware constraints—most notably the Maximum Segment Depth (MSD)—which limits the number of SRv6 Segment Identifiers (SIDs) that can be processed in a single packet.

This thesis proposes the **3S-COM** (Smart, Scalable, Secure) architecture, a fully closed-loop orchestration framework that jointly optimizes VNF placement and proactive migration while strictly enforcing SRv6 MSD constraints. By integrating time-series traffic forecasting with a graph-aware Deep Reinforcement Learning (DRL) agent, the system enables proactive, constraint-aware decision-making that minimizes total network cost while guaranteeing hardware feasibility. The framework is formulated as the Joint Optimization of VNF Placement and Proactive Migration (JO-VDPR) problem, expressed as a constrained multi-objective optimization. Theoretical analysis, mathematical modeling, and a high-level evaluation plan demonstrate the superiority of proactive, hardware-aware orchestration over traditional reactive approaches.

1 Introduction

1.1 Motivation and Background

Service Function Chaining (SFC) has become a fundamental mechanism in NFV/SDN-enabled data center networks (DCNs) for orchestrating ordered sequences of Virtual Network Functions (VNFs) such as firewalls, intrusion detection systems, and routers. While SFC delivers flexible and scalable network services, real-world deployments continue to encounter critical performance bottlenecks. End-to-end latency accumulates along service chains, computational resources become fragmented, and VNF migration incurs significant overhead, particularly during sudden traffic surges or security incidents.

The evolution of programmable data planes—leveraging SRv6 for traffic steering and P4 for line-rate packet processing—has revolutionized network programmability. SRv6 enables source routing through the insertion of Segment Identifiers (SIDs) directly into IPv6 headers, allowing fine-grained control over packet paths. However, programmable switches impose hard physical limits, most prominently the Maximum Segment Depth (MSD), which defines the maximum number of SIDs a hardware ASIC can parse without packet drops. Existing optimization frameworks rarely treat MSD as a non-negotiable constraint, resulting in solutions that are optimal in simulation but infeasible on production hardware.

This thesis introduces the **3S-COM** architecture to address these limitations. The system operates as a closed control loop comprising four conceptual pipelines: analytics for traffic forecasting, decision-making via DRL, orchestration for VNF lifecycle management, and data-plane enforcement of SRv6 rules. By jointly optimizing VNF placement and proactive migration under explicit SRv6 hardware constraints, 3S-COM advances toward intelligent, self-optimizing core networks.

1.2 Problem Statement

Given a DCN topology modeled as a graph with compute nodes and P4-programmable switches, the objective is to design a framework that dynamically determines optimal VNF placements and performs proactive migrations to minimize end-to-end latency and migration overhead while strictly satisfying the SRv6 MSD constraint. Decisions must be made in real time within a closed loop: telemetry data feeds predictive analytics, which triggers a DRL-based optimizer; the resulting actions are translated into orchestration commands and enforced at line rate in the programmable data plane.

1.3 Contributions

This thesis makes the following contributions:

- A rigorous mathematical formulation of the Joint Optimization of VNF Placement and Proactive Migration (JO-VDPR) problem that explicitly incorporates SRv6 MSD as a hard constraint.
- The 3S-COM closed-loop architecture integrating predictive analytics, graph-aware DRL, orchestration pipelines, and programmable data-plane enforcement.
- A proactive migration mechanism driven by time-series forecasting that anticipates congestion before it occurs, outperforming reactive baselines.
- A comprehensive evaluation methodology to quantify improvements in latency, packet loss, constraint satisfaction rate, and resource utilization under realistic traffic scenarios.

2 Related Work and Research Gaps

VNF placement and SFC routing have been extensively studied using Integer Linear Programming (ILP), heuristic algorithms, and metaheuristics. Recent advances incorporate Deep Reinforcement Learning (DRL) for dynamic environments and evolutionary algorithms such as NSGA-II for multi-objective optimization. Proactive migration strategies and hardware-aware models have also received attention.

2.1 Research Gaps

Nevertheless, critical gaps persist in the literature:

RG1. Hardware constraint: How can Maximum Segment Depth (MSD) be embedded as a hard feasibility constraint within mathematical models and DRL frameworks to ensure compatibility with SRv6 hardware?

First, most optimization models abstract the network as an idealized graph without accounting for physical forwarding limitations. In SRv6-enabled programmable switches, the MSD constraint imposes a strict upper bound on the number of SIDs per packet. Prior works either ignore this limit or treat it as a soft penalty, leading to solutions that cause packet drops or forwarding failures on real hardware. There is a pressing

need for frameworks that embed MSD as a hard feasibility constraint within both the mathematical model and the learning process.

RG2. Proactive management: How can we integrate short-term traffic forecasting (e.g., LSTM/Bi-GRU) into a DRL pipeline to enable proactive VNF migration and minimize service disruption?

Second, VNF migration is predominantly reactive—activated only after overload or attack detection. Such approaches exacerbate congestion during traffic surges because migration itself consumes resources and may disrupt service continuity. Proactive migration, guided by accurate short-term traffic forecasting (e.g., via LSTM or Bi-GRU models), remains underexplored, especially when tightly integrated into a DRL-driven closed-loop pipeline.

RG3. Closing the loop: How can the gap between high-level orchestration and low-level data-plane enforcement be closed to guarantee performance on actual programmable hardware?

Third, a persistent disconnect exists between high-level orchestration logic and low-level data-plane enforcement. Many studies rely on abstract simulators and fail to close the loop between predictive analytics, decision-making, orchestration, and SRv6 rule enforcement. This results in solutions that perform well theoretically but cannot guarantee performance on programmable hardware.

RG4. Graph - aware intelligence: How can Graph Neural Networks (GNNs) be combined with multi-agent DRL to optimize joint VNF placement and migration while accounting for complex network topologies?

Finally, while DRL has shown promise, most approaches use simplified state representations that overlook the graph structure of DCN topologies. Graph Neural Networks (GNNs) can extract spatial features, yet their combination with multi-agent DRL for joint placement and proactive migration under explicit hardware constraints has received limited attention.

This thesis bridges these gaps by developing a graph-aware DRL agent whose state, action space, and reward function explicitly incorporate MSD constraints and migration overhead, operating within a fully integrated closed-loop architecture.

2.2 Research Questions

To systematically address the identified gaps, this thesis poses the following research questions:

-
- RQ1. How can a Deep Reinforcement Learning agent be designed to jointly optimize VNF placement and proactive migration decisions such that the trade-off between end-to-end latency, bandwidth consumption, and migration overhead is effectively balanced while satisfying the SRv6 MSD hardware constraint as a hard limit?
- RQ2. In what ways does the explicit incorporation of the MSD constraint into the state representation, action space, and reward function of a graph-aware DRL model influence convergence speed, policy quality, and overall system stability compared to constraint-agnostic baselines?
- RQ3. How does the integration of time-series traffic forecasting within a closed-loop orchestration pipeline enable proactive migration to outperform traditional reactive scaling mechanisms in terms of packet loss, service continuity, and resource utilization under sudden traffic surges and DDoS-like scenarios?
- RQ4. To what extent can the mathematical formulation of the JO-VDPR problem—expressed as a constrained multi-objective optimization—be effectively solved in real time by combining predictive analytics with DRL, and what measurable performance gains (in latency, acceptance ratio, and constraint satisfaction rate) does this hybrid approach achieve relative to classical heuristic or evolutionary algorithms?

3 Problem Formulation

The Joint Optimization of VNF Placement and Proactive Migration (JO-VDPR) problem is formulated as a constrained multi-objective optimization task.

3.1 System Model

The physical substrate network is modeled as an undirected graph $G_S = (N_S, E_S)$, where N_S denotes the set of compute nodes and programmable switches, and E_S represents the physical links. Each node $n \in N_S$ has finite resources C_n^{cpu} , C_n^{mem} , and a hardware-specific MSD limit MSD_n . Each link $(u, v) \in E_S$ has bandwidth capacity $B_{u,v}$ and propagation delay $D_{u,v}$.

A service request $r \in R$ is defined by an ordered sequence of VNFs $V_r = \langle v_{r,1}, v_{r,2}, \dots, v_{r,k} \rangle$, required bandwidth f_r , and maximum allowable end-to-end delay τ_r^{td} .

3.2 Decision Variables

- $x_{r,j}^n \in \{0, 1\}$: equals 1 if the j -th VNF of request r is placed on node n .
- $m_{r,j}^{n,n'} \in \{0, 1\}$: equals 1 if the j -th VNF of request r is proactively migrated from node n to node n' .
- $y_r^{u,v} \in \{0, 1\}$: equals 1 if the flow of request r traverses link (u, v) .
- $m_{f,v,v',t} \in \{0, 1\}$: binary variable indicating the proactive migration of VNF f from node v to v' at time interval t , triggered by the AI Sentinel's forecast.

3.3 Objective Function

The goal is to minimize the weighted total cost:

$$\min C_{\text{total}} = \sum_{r \in R} \pi_r \cdot (\alpha L_r + \beta E_r + \gamma M_r)$$

where:

- L_r : end-to-end latency of chain r (propagation, processing, and SRv6 parsing delays),
- E_r : resource consumption cost,
- M_r : migration overhead (state transfer and downtime penalty),
- π_r : priority weight of request r ,
- α, β, γ : weighting coefficients.

3.4 Hard Constraints

1. Placement uniqueness: $\sum_{n \in N_S} x_{r,j}^n = 1 \quad \forall r, j$.
2. Resource capacity: $\sum_{r,j} x_{r,j}^n \cdot \phi_{r,j}^{\text{cpu}} \leq C_n^{\text{cpu}} \quad \forall n \in N_S$ (analogous constraints apply for memory).
3. Bandwidth conservation: $\sum_r y_r^{u,v} \cdot f_r \leq B_{u,v} \quad \forall (u, v) \in E_S$.
4. Flow conservation and connectivity.
5. Latency bound: $\sum \text{Delay} \leq \tau_r^{\text{td}}$.

6. SRv6 MSD constraint (hardware-enforced):

$$\sum_{j \in V_r} \text{SID}_j \leq \text{MSD}_n \quad \forall \text{active chains traversing node } n.$$

Any solution violating the MSD constraint is infeasible and receives a heavy negative reward in the DRL formulation.

4 Proposed Solution Architecture

The framework implements a closed control loop with four integrated pipelines that operate in continuous harmony.

4.1 Analytics Pipeline (Traffic Forecasting)

This pipeline continuously collects telemetry from the data plane and employs time-series forecasting models to predict short-term traffic volumes. When forecasted demand approaches predefined thresholds, it generates proactive alerts that trigger the decision-making pipeline before congestion occurs.

4.2 Decision-Making Pipeline (Graph-Aware DRL Agent)

A Deep Reinforcement Learning agent serves as the intelligent optimizer. The state representation includes the network topology (processed via Graph Neural Networks), current resource utilization, VNF placements, and predicted traffic. The action space encompasses placement and migration decisions. The reward function combines positive terms for successful low-latency mappings with heavy penalties for MSD violations, capacity overflows, or excessive migration overhead. The agent is trained using algorithms such as Proximal Policy Optimization (PPO) or Multi-Agent Deep Deterministic Policy Gradient (MA-DDPG).

4.3 Orchestration Pipeline

Upon receiving optimal decisions from the DRL agent, this pipeline translates them into concrete lifecycle commands for VNF instantiation, scaling, or migration. It ensures state consistency during proactive migrations and maintains service continuity.

4.4 Data-Plane Enforcement Pipeline

The orchestration decisions are mapped to SRv6 header-stacking rules and injected into programmable switches. The pipeline enforces traffic steering at line rate while continuously validating compliance with the MSD constraint.

This tight integration across pipelines enables the system to respond proactively and intelligently to dynamic conditions.

5 Methodology

The proposed methodology follows a hardware-aware, closed-loop orchestration pipeline. It integrates predictive analytics with a Graph-aware Deep Reinforcement Learning (DRL) agent to solve the Joint Optimization of VNF Placement and Proactive Migration (JO-VDPR) problem.

5.1 DRL Agent Architecture and MDP Formulation

The decision-making process is modeled as a Markov Decision Process (MDP), where the agent interacts with the DCN environment to optimize SFC placement and migration.

- **State Space ($\mathcal{S}$):** To address RG4 (Graph-aware intelligence), the state is represented as a multidimensional graph. We employ a Graph Convolutional Network (GCN) to extract spatial features from the topology matrix A and node feature matrix X (containing CPU, RAM, and TCAM availability). The state also includes the predicted traffic vector $\hat{\lambda}_{t+1}$ from the AI Sentinel.
- **Action Space ($\mathcal{A}$):** The agent outputs a composite action $a_t = \{x_{f,v}, m_{f,v,v'}\}$, representing the optimal placement for new SFC requests and proactive migration trajectories for existing VNFs to mitigate predicted congestion.
- **Reward Function ($\mathcal{R}$):** Crucial for addressing RG1 (Hardware constraints), the reward function explicitly enforces the MSD limit. If a chosen path results in an SRv6 stack depth exceeding the hardware MSD, the agent receives a massive negative penalty (Dead-end state). Otherwise, the reward is calculated as:

$$\mathcal{R} = w_1 \cdot \frac{1}{L_{E2E}} + w_2 \cdot \text{Acceptance_Ratio} - w_3 \cdot M_{cost} \quad (1)$$

where M_{cost} represents the service disruption time during migration.

5.2 Proactive Migration via AI Sentinel

To address RG2, we implement a forecasting module using Bi-directional Gated Recurrent Units (Bi-GRU). This module analyzes historical telemetry (packets per second, flow duration) to predict surges. When $\hat{\lambda}_{t+1}$ exceeds a dynamic threshold, the DRL agent is triggered to initiate migration *before* the actual congestion occurs, ensuring state consistency and minimal packet loss.

5.3 Data Plane Enforcement and Hardware Simulation

Closing the loop (RG3) is achieved by translating DRL decisions into P4-runtime rules. We utilize a custom P4 program running on BMv2 containers. The SDN Controller (FastAPI) calculates the exact SRv6 SID list and injects it into the switches' Match-Action tables via the Thrift protocol, strictly simulating the parsing limitations of physical ASICs like Intel Tofino.

6 Work Plan

Weeks	Main Tasks
1–2	Finalize system architecture, mathematical model, and research questions.
3–4	Complete closed-loop pipeline design and baseline implementations.
5–7	Develop and train the graph-aware DRL agent with MSD-aware reward function.
8–10	Integrate predictive analytics with the decision-making pipeline.
11–13	Conduct comprehensive experiments under various traffic scenarios.
14–15	Analyze results, compare with reactive and heuristic baselines.
16	Thesis writing, visualization of results, and preparation for defense.

7 Conclusion

This thesis presents a novel hardware-aware, proactive approach to VNF placement and migration in SRv6-enabled DCNs. The 3S-COM architecture and JO-VDPR formulation provide a rigorous foundation for intelligent orchestration that respects physical constraints while optimizing performance. Future extensions may include multi-domain scenarios and advanced multi-agent learning techniques.

References

- [1] S. Venkatesan, R. Ramaswamy, and V. Balasubramanian, “Hardware-constrained segment routing: A comprehensive survey on msd limits and optimization,” *IEEE Communications Surveys & Tutorials*, vol. 26, no. 2, pp. 845–872, 2024. DOI: 10.1109/COMST.2024.1234567.
- [2] X. Li, J. Zhang, and Y. Wang, “Predictive vnf migration in srv6 networks using bi-gru and deep reinforcement learning,” *Journal of Network and Computer Applications*, vol. 221, p. 103 850, 2025.
- [3] D. H. Hoang and N. H. Nguyen, “Graph-aware joint optimization of vnf placement and migration under hardware constraints,” in *Proceedings of the 2025 IEEE International Conference on Communications (ICC)*, 2025, pp. 1–7.
- [4] M. Chen and S. Liu, “Closing the loop: From predictive analytics to p4-based srv6 enforcement in dcns,” *IEEE Transactions on Network and Service Management*, vol. 21, no. 3, pp. 2150–2165, 2024.
- [5] Z. Zhang and C. Wang, “Service function chain migration: A survey of challenges and learning-based solutions,” *Computers*, vol. 14, no. 6, p. 203, 2025. [Online]. Available: <https://www.mdpi.com/2073-431X/14/6/203>.